Oct. 7

Ch 7

~~Real~~
Recall: Harmonic function is a f'n that satisfies the Laplace eq'n over a domain, Ω, & is $C^2(\Omega)$

Laplace Eq'n, $\Delta u = 0$ $\qquad \Delta = \underline{\nabla} \cdot \underline{\nabla} = \underline{\nabla}^2$

In cartesian coord.

$n=2 \quad u_{xx} + u_{yy} = 0$

$n=3 \quad u_{xx} + u_{yy} + u_{zz} = 0$

$(x,y) \to (r,\theta)$

$x = r\cos\theta \quad y = r\sin\theta$

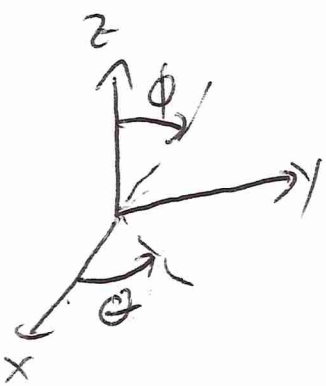
$x^2+y^2 = r^2$

$\tan\theta = \frac{y}{x}$

In polar coord

$n=2 \quad \Delta u = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2} = 0$

$\frac{1}{L} \quad \frac{1}{L} \quad L \quad \frac{1}{L}$



$n \geq 3 \quad \Delta u = \frac{1}{r^{n-1}}\frac{\partial}{\partial r}\left(r^{n-1}\frac{\partial u}{\partial r}\right) + \frac{1}{r^2}\Delta_n u = 0$

$\Delta_3 u = \frac{1}{\sin\phi}\frac{\partial}{\partial \phi}\left(\sin\phi\frac{\partial u}{\partial \phi}\right) + \frac{1}{\sin^2\phi}\frac{\partial^2 u}{\partial \theta^2}$

$=$ Laplace - Beltramic operator

Examples of simple harmonic f'ns ($\Delta u = 0$)

$n=2$ $\quad u = \ln r$ (textbook $\ln \to \log r$, $\log_e r$)

$n \geq 3$ $\quad u = \frac{1}{r^{n-2}}$ note how 2 dim. harmonic f'n can be quite different than harmonic ~~fn~~ f'ns for $n \geq 3$.

§7.2 How to find other harmonic f'ns.

Separation of Variables

$n=2$ polar coord

$$\Delta u = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2} = 0$$

Look for a sol'n of the form $u(r,\theta) = R(r)\Theta(\theta)$

not all f'ns $u(r,\theta)$ can be written as $R(r)\Theta(\theta)$

e.g. $r+\theta$ $\quad r^\theta$

Analogous to solving $a\ddot{x} + b\dot{x} + cx = 0$ Assume $x = e^{\lambda t}$

It works! + existence + uniqueness Thms tell us it doesn't matter how we get to the sol'n.

$u(r,\theta) = R(r)\Theta(\theta)$

subst. into eq'n

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial(R(r)\Theta(\theta))}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2(R(r)\Theta(\theta))}{\partial\theta^2} = 0$$

$$\Theta(\theta)\frac{1}{r}\underbrace{\frac{\partial}{\partial r}(rR'(r))}_{\text{product rule}} + \frac{R(r)}{r^2}\frac{\partial^2\Theta(\theta)}{\partial\theta^2} = 0$$

$$R'(r) = \frac{dR}{dr} \qquad \Theta'(\theta) = \frac{d\Theta}{d\theta}$$

$$\Theta(\theta)\frac{1}{r}\left[R'(r) + rR''(r)\right] + \frac{R(r)}{r^2}\Theta''(\theta) = 0$$

separate:

$$\frac{r^2}{R\,\Theta}\left[\Theta(\theta)\frac{1}{r}\left[R' + rR''\right] = -\frac{R}{r^2}\Theta''\right]$$

$$\underbrace{\frac{r}{R}\left[R' + rR''\right]}_{\substack{\text{f'n only depends}\\ \text{on } r}} = \underbrace{-\frac{1}{\Theta}\Theta''}_{\substack{\text{f'n that only}\\ \text{depends on } \theta}} = \mu$$

↑ constant

$$f(r) = g(\theta)$$

Solve for $R(r)$:

$$\frac{r^2 R'' + r R'}{R} = \mu$$

$$r^2 R'' + r R' - \mu R = 0$$

<u>case 1</u> $\mu \neq 0$

Assume $R = r^\alpha$ $\quad R' = \alpha r^{\alpha-1}$ $\quad R'' = \alpha(\alpha-1) r^{\alpha-2}$

subst. in

$$r^2\left[\alpha(\alpha-1) r^{\alpha-2}\right] + r\alpha r^{\alpha-1} - \mu r^\alpha = 0$$

solve for α

$$\alpha(\alpha-1) r^\alpha + \alpha r^\alpha - \mu r^\alpha = 0$$

$$r^\alpha\left[\alpha^2 - \cancel{\alpha} + \cancel{\alpha} - \mu\right] = 0$$

$r^\alpha \neq 0$ Not a sol'n over domain

$$\alpha = \pm\sqrt{\mu}$$

$$R(r) = r^{\pm\sqrt{\mu}}$$

In general need to consider $\mu > 0$, $\mu < 0$, $\mu = 0$

$\mu < 0 \Rightarrow$ complex sol'ns + we only want real sol'ns, so Throw out.

<u>case 2</u> $\mu = 0$

$$r^2 R'' + r R' = 0$$

Let $S = R'$

$$r^2 S' + r S = 0$$

Sol! $r^2 \frac{dS}{dr} = -rS$

$$\frac{dS}{S} = -\frac{1}{r}\,dr$$

$$a \ln b = \ln(b^a)$$

$$\ln S = -\ln r + c_1$$

$$S = e^{-\ln r + c_1} = c_1 e^{\ln(r^{-1})} = \frac{c_1}{r}$$

$$R' = \frac{c_1}{r}$$

$$R = c_1 \ln r + c_2$$

$\overbrace{\{\ln r, 1\}}^{\text{"basis" or fund. sol'n set for}}$ $r^2 R'' + r R' = 0$

$$R_\mu(r) = \begin{cases} 1, \ \ln r & \text{if } \mu = 0 \\ r^{\sqrt{\mu}}, \ r^{-\sqrt{\mu}} & \text{if } \mu > 0 \end{cases}$$